

MHF 4U1- Unit 5 Quiz AM - Sinusoidal Functions

Name _____

Date _____

Total

17

/17

Circle the correct answer for questions 1 - 5.

1. Given the trigonometric function $y = \tan x$, find the x-coordinate of the point $(?, 0)$?

a. $\frac{\pi}{2}$

b. $\frac{\pi}{4}$

c. $\frac{\pi}{3}$

d. π ✓

2. The height, h , in metres, above the ground of a car as a Ferris wheel rotates can be modelled by the function $h(t) = 16\cos\left(\frac{\pi t}{120}\right) + 18$, where t is the time in seconds. What is the maximum height the car?

a. 18m

b. 26m

c. 34m ✓

d. 17m

3. Determine the phase shift of the sinusoidal function $y = -3\sin\left(2x - \frac{\pi}{6}\right) + 1$. $\rightarrow y = -3\sin\left[2\left(x - \frac{\pi}{12}\right)\right] + 1$

a. $\frac{\pi}{6}$ rad to the left

b. $\frac{\pi}{6}$ rad to the right

c. $\frac{\pi}{3}$ rad to the right

d. $\frac{\pi}{12}$ rad to the right ✓

4. Give an equation for the transformed sine function with amplitude of 3, a period of $\frac{\pi}{4}$, and a phase shift of $\frac{\pi}{5}$ rad to the right.

a. $y = 3\sin\left[\frac{\pi}{4}\left(t - \frac{\pi}{5}\right)\right]$

b. $y = 3\sin\left[8\left(t - \frac{\pi}{5}\right)\right]$ ✓

c. $y = 3\sin\left[\frac{\pi}{4}\left(t + \frac{\pi}{5}\right)\right]$

d. $y = 3\sin\left[8\left(t + \frac{\pi}{5}\right)\right]$

$\hookrightarrow k = \frac{2\pi}{\frac{\pi}{4}} = 8$

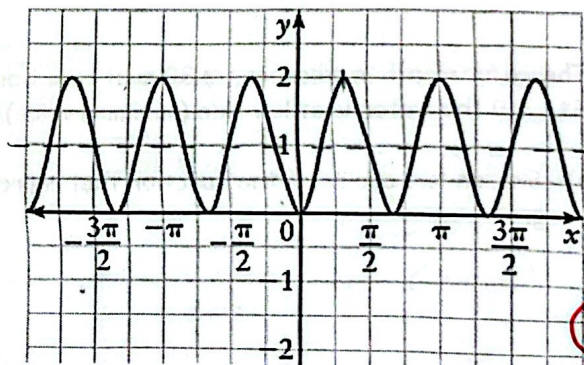
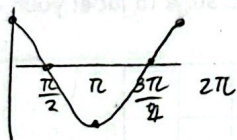
5. Determine an equation for the sinusoidal function shown.

a. $y = -2\cos x + 1$

b. $y = -\cos \pi x + 1$ ✓

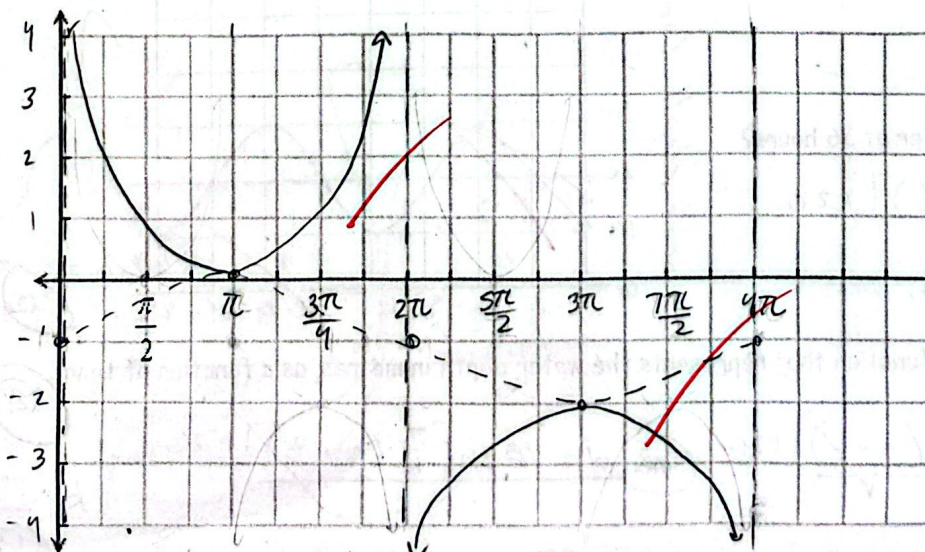
c. $y = 2\cos \pi x$

d. $y = -2\cos x$



(5)

6. Sketch the graph of $f(x) = \csc \frac{1}{2}x - 1$ for $0 \leq x \leq 4\pi$.



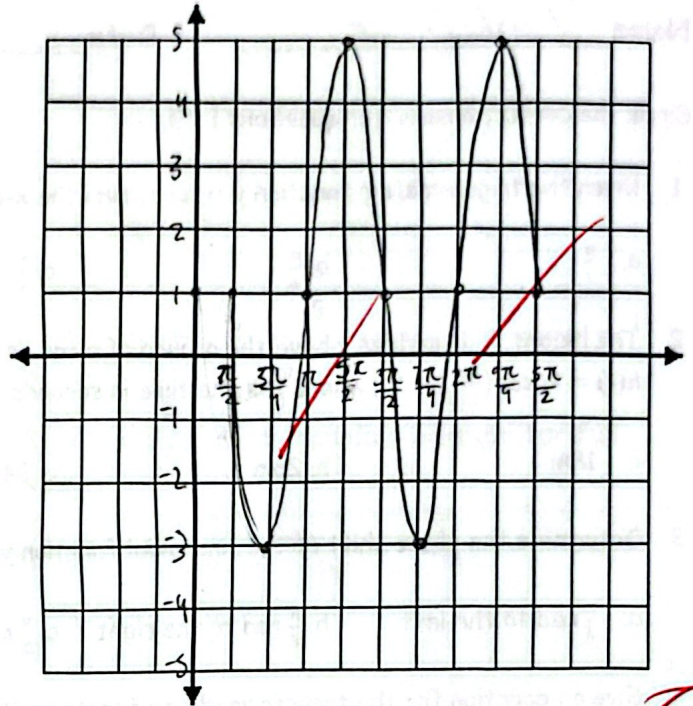
$\sin \theta = (2x - 1)$
 $(0, 0) \rightarrow (0, -1)$
 $(\frac{\pi}{2}, 1) \rightarrow (\pi, 0)$
 $(\pi, 0) \rightarrow (2\pi, -1)$
 $(\frac{3\pi}{2}, -1) \rightarrow (3\pi, 0)$
 $(2\pi, 0) \rightarrow (4\pi, -1)$

solid line is $\csc \frac{1}{2}x - 1$

(2)

7. a) Sketch two cycles of $f(x) = -4\sin\left[2\left(x - \frac{\pi}{2}\right)\right] + 1$. You can use the mapping rule or 5-point method to sketch. Show all your work!

$\sin x \rightarrow \left(\frac{1}{2}x + \frac{\pi}{2}, -4y + 1\right)$
 $(0, 0) \rightarrow \left(\frac{\pi}{2}, 1\right)$
 $\left(\frac{\pi}{2}, 1\right) \rightarrow \left(\frac{3\pi}{4}, -3\right)$
 $(\pi, 0) \rightarrow (\pi, 1)$
 $\left(\frac{3\pi}{2}, -1\right) \rightarrow \left(\frac{5\pi}{4}, 5\right)$
 $(2\pi, 0) \rightarrow \left(\frac{3\pi}{2}, 1\right)$
 $\left(\frac{5\pi}{2}, 1\right) \rightarrow \left(\frac{7\pi}{4}, -3\right)$
 $(3\pi, 0) \rightarrow (2\pi, 1)$
 $\left(\frac{7\pi}{2}, -1\right) \rightarrow \left(\frac{9\pi}{4}, 5\right)$
 $(4\pi, 0) \rightarrow \left(\frac{3\pi}{2}, 1\right)$



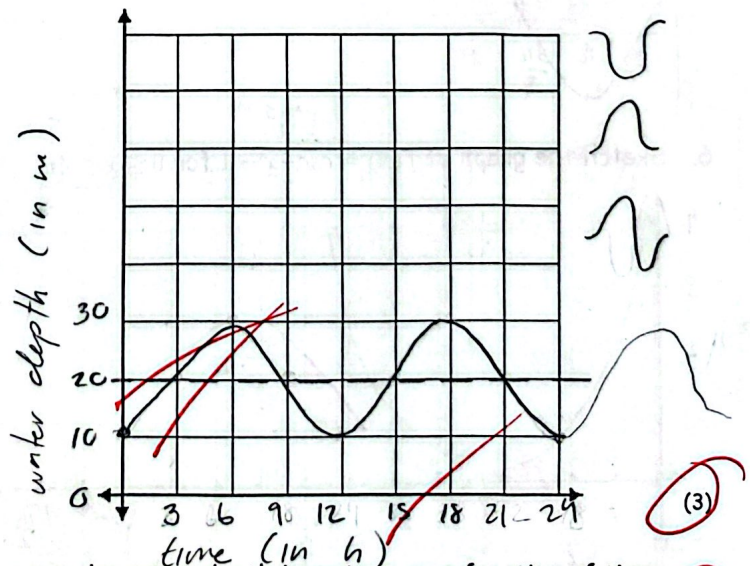
- b) State the following information about the above graph.

Amplitude: 4 Period: π Phase shift: $\frac{\pi}{2}$ right Range: $\{y \in \mathbb{R} \mid -3 \leq y \leq 5\}$

8. The water depth in a harbour is 30 m at high tide and 10 m at low tide. One cycle is completed every 12 hours. At $t = 0$ the water is at low tide (minimum value).

- a) Sketch two cycles of the function that represents the water depth in the harbour. Be sure to label your graph.

$(6, 20)$



- b) What is the depth of the water at 36 hours?

$f(x) = 10 \cos\left[\frac{\pi}{6}(36 - 6)\right] + 10$
 $= 10 \text{ m}$
 $(36, 10) \text{ @ } 36 \text{ h, } 10 \text{ m}$

- c) Determine a **cosine** and **sine** function that represents the water depth in metres, as a function of time.

Cosine: $d = 10 \cos\left[\frac{\pi}{6}(x - 6)\right] + 20$

Sine: $d = 10 \sin\left[\frac{\pi}{6}(x - 3)\right] + 20$

$k = \frac{2\pi}{12}$
10